A Multimodal Framework for Electricity Price Forecasting Using Sentiment-Augmented Time-Series Signals

Rohan Fozdar

Knox College

Abstract

This paper presents a multimodal forecasting framework for real-time electricity prices on the Electric Reliability Council of Texas (ERCOT) grid, spanning approximately 80,000 hourly observations from 2016 through 2024. Six data sources are used: historical prices, system load, weather, natural gas market data, nuclear reactor outage status, and news sentiment derived from the GDELT 2.0 corpus. Against this shared data infrastructure, twelve forecasting architectures are evaluated under identical train/validation/test conditions to forecast next-hour price. Model families include gradient boosted trees, deep learning models, zero-shot foundation models, classical time series methods, and hybrid approaches. Results show that sentiment augmentation via GDELT improves mean absolute error by approximately 10%.

1. Introduction

Electricity price forecasting is one of the harder regression problems in applied machine learning. Unlike commodity markets, wholesale electricity prices are set in real time. They respond almost immediately to changes in production, demand, costs, and weather-change. Because of this, even small forecasting errors can have severe consequences.

ERCOT, the Texas electricity market, is an especially challenging case. It operates as a largely isolated grid serving roughly 90% of Texas electric load, with limited ability to import power from neighboring regions. Its competitive wholesale market has also produced some of the most extreme price events in North American electricity markets (Potomac Economics, 2021). In the 2016–2024 period studied here, the hub-average real-time price had a median of about $30/MWh, but during grid stress events prices could surge toward the $9,000/MWh regulatory cap. About 2.5% of hourly observations exceeded $200/MWh, while roughly 1% were negative, often during periods of high renewable generation and low demand. This creates an unusually difficult target distribution, with skewness above 10 and kurtosis above 100.

This paper makes three main contributions. First, it builds a feature matrix from nine years of hourly ERCOT data, combining six signal families: prices, load, weather, natural gas, nuclear reactor status, and news sentiment. Second, it compares 12 forecasting models across different families under the same chronological splits and evaluation metrics. The models are tested on predicting next-hour price in $/MWh. Third, it shows that GDELT-derived news sentiment provides consistent improvements over a physical-feature-only baseline.

2. Data and Features

2.1 Data Sources

The feature matrix is assembled from six independent signal families, each retrieved from a distinct provider and joined on a shared UTC hourly timestamp.

Price data is sourced directly from ERCOT public market archives and comprises both real-time and day-ahead settlement prices at five hub locations: HB_HUBAVG (the grid-wide hub average, which serves as the regression target), HB_HOUSTON, HB_NORTH, HB_SOUTH, and HB_WEST. Day-ahead prices for the same five hubs are included as forward-looking features; because day-ahead markets clear the evening before delivery, these prices are observable at forecast time without risking future leakage. System load, representing grid-wide electricity demand in megawatts, is drawn from ERCOT's hourly load reports.

Weather observations are retrieved from the Open-Meteo historical API for four Texas cities : Houston, Dallas (North), San Antonio (South), and Midland (West); covering temperature, relative humidity, cloud cover, wind speed, wind gusts, and precipitation. Natural gas market data, including the Henry Hub spot price, storage levels in billion cubic feet, and week-over-week storage changes, originates from the U.S. Energy Information Administration (EIA); these series are sourced from EIA Excel files available on their website.

Nuclear reactor outage data is collected from the Nuclear Regulatory Commission's daily reactor status reports, with per-unit outage percentages and aggregate offline capacity computed for each of the 4 reactors.

News sentiment is derived from the GDELT website which offers users the ability to get news sentiment scores on the searched keyword. We pulled the average tone, normalized tone, and article volume fields.

2.2 Feature Engineering

Raw signals are joined by timestamp and passed through several engineering steps that produce the 87-feature matrix covering 77,956 hourly observations from January 2016 through December 2024.

Price history features are constructed from HB_HUBAVG using lags at 1, 12, 24, 48, and 168 hours, along with rolling mean and standard deviation at 24-hour and 168-hour windows. The 720-hour lag horizon was considered, but not implemented because it requires discarding the first 720 observations, bringing the effective training start to late January 2016. Day-ahead hub prices are lagged by one day per hub. Temporal calendar features include hour of day, day of week, month, a weekend indicator, and a peak-hours indicator (weekday 16:00–22:00 local time).

The Binary grid-stress flags introduced were: stress_cold (<32°F), stress_freeze (<0°F), stress_heat (>90°F), stress_extreme_heat (>105°F), stress_low_wind, stress_gas_spike, and stress_reactor_outage. All these flags are combined into a composite stress_score [0,7]. Weather observations are aggregated across the four city zones into cross-zone maximum, minimum, and mean for temperature and wind.

Feature transformations were applied to select features that showed a heavy right-skew in residuals. Log(1+x) or Log1p was applied to: price lag and rolling columns (10 columns), per-city wind speed and gust columns (8 columns), day-ahead hub price columns (5 columns), per-city precipitation columns (4 columns), and natural gas price and storage columns (4 columns). In total, 34 of the 87 features are log-transformed; the remaining 53 pass through unchanged. This transformation table was derived from residual diagnostics described in Section 4.

2.3 Target and Split

One target is defined. The regression target is HB_HUBAVG in $/MWh, evaluated on mean absolute error (MAE) and root mean squared error (RMSE).

All models use an identical split with no random shuffling: training on 2016–2022 (60,428 observations), validation on 2023 (8,760 observations), and testing on 2024 (8,767 observations). All reported metrics are computed on the 2024 test set.

3. Models and Evaluation Methodology

3.1 Evaluation Design

All sixteen models are evaluated under identical conditions: the same chronological split, the same 87-feature matrix after applying feature transformations, and the same metric functions. This design ensures that differences in reported metrics are attributable to model choice rather than data preparation.

3.2 Model Taxonomy

The twelve models span five families:

Gradient Boosted Trees (GBT): XGBoost, LightGBM, CatBoost, and NGBoost.

Deep Learning: a bidirectional LSTM with a 168-hour look-back window, and a Temporal Fusion Transformer.

Foundation Models (zero-shot, univariate): Chronos T5-small (Ansari et al., 2024) with a 512-hour context window, and TimesFM 2.5-200M (Das et al., 2024) with a 512-hour context window batched over the test set. Zero Shot means that both models receive only the price series without exogenous features.

Classical Time Series: SARIMAX with daily seasonality (m=24) plus all exogenous regressors, and Meta's Prophet (Taylor & Letham, 2018) with an all-87-feature configuration.

Hybrid Models: Two foundation-model-augmented XGBoost hybrids in which the foundation forecast is appended as an additional feature column before XGBoost training.

Model	MAE ($/MWh)	RMSE ($/MWh)	Spike Recall	Spike Precision
XGBoost (No sentiment)	$34.53/MWh	$122.57/MWh	50.53%	19.54%
XGBoost	$24.43/MWh	$112.46/MWh	57.02%	25.10%
LightGBM	$19.76/MWh	$100.41/MWh	34.21%	45.35%
CatBoost	**$13.75/MWh**	**$95.48/MWh**	**44.35%**	**56.94%**
NGBoost	$14.59/MWh	$106.44/MWh	25.44%	59.18%
LSTM	$24.42/MWh	$148.72/MWh	16.67%	23.46%
TFT	$45.75/MWh	$134.60/MWh	0.00%	0.00%
CHRONOS (amazon)	**$14.72/MWh**	**$115.59/MWh**	**42.11%**	**60.76%**
PROPHET (meta)	$119.14/MWh	$168.75/MWh	57.14%	17.70%

TIMESFM (google)	$32.51/MWh	$136.94/MWh	1.75%	11.76%
SARIMAX	$18.45/MWh	$124.31/MWh	45.61%	38.81%
CHRONOS Hybrid	**$12.13/MWh**	**$93.98/MWh**	**61.40%**	**50.43%**
PROPHET Hybrid	$48.36/MWh	$108.96/MWh	71.05%	14.16%

4. Results

4.1 Gradient Boosted Trees

The XGBoost baseline, trained on all 87 features including GDELT sentiment, achieves a test MAE of $24.43/MWh and RMSE of $112.46/MWh, with spike recall of 57.02% and spike precision of 25.10%. This model serves as the primary reference point for the comparison.

Among the GBT family, CatBoost achieves the lowest MAE of $13.75/MWh and RMSE of $95.48/MWh, outperforming XGBoost on both metrics. LightGBM follows closely with MAE of $19.76/MWh and RMSE of $100.41/MWh. NGBoost, which predicts a full Normal distribution over the price rather than a point estimate, achieves MAE of $14.59/MWh at the cost of higher RMSE ($106.44/MWh) and substantially reduced spike recall (25.44%), consistent with its probabilistic objective smoothing over the spike tail. Across the GBT family, there is a consistent pattern: models that minimize MAE, do so by becoming more conservative on spike hours, trading recall for precision.

Catboost here performs the best because of Ordered Boosting. It differs from the other members of the family in the way gradient is computed by the model. Traditional GBT models are given access to all data points when computing gradients, which is in essence, future leakage. This is why models perform very well in train and val, but during test, when they do not have access to the future values in gradient computation, their performance drops significantly. CatBoost overcomes this weakness since it is never given access to future values during gradient computation. On trading day N, the model only has access to data from 1 to (N-1). This is why it outperforms other models on the test set.

4.2 Deep Learning

The bidirectional LSTM with a 168-hour look-back window achieves MAE of $24.42/MWh, nearly identical to the XGBoost baseline, but with substantially higher RMSE of $148.72/MWh. This gap between MAE and RMSE implies that the LSTM handles typical hours comparably to XGBoost but performs significantly worse on spike hours. Spike recall of 16.67% confirms that the LSTM is largely failing to anticipate extreme price events, likely because the architecture does not have enough data points to train on. 80,000 rows is a lot of data for GBT models, but inconsequential to traditional deep learning frameworks

The Temporal Fusion Transformer performs poorly in this configuration, with MAE of $45.75/MWh, RMSE of $134.60/MWh, and zero spike recall or precision. The TFT's additive attention mechanism is well-suited to problems where temporal patterns are the primary driver of variation, but ERCOT prices are driven heavily by nonlinear interactions between physical state variables like load, weather, fuel costs, and generation mix, that tree-based methods handle more naturally through learned splits. The zero spike recall suggests the TFT's outputs are not producing predictions that cross the $200/MWh threshold at any point in the test year, indicating the model has not learned the conditions under which extreme prices arise, solidifying our claim that deep learning models do not have enough data to train on in the electricity space.

4.3 Foundation Models and Hybrids

The zero-shot foundation models operate without any exogenous features. Chronos T5-small, which receives only the univariate price series with a 512-hour context window, achieves MAE of $14.72/MWh but with RMSE of $115.59/MWh, suggesting it handles typical hours well while struggling on spikes. TimesFM 2.5-200M performs considerably worse, with MAE of $32.51/MWh and near-zero spike recall (1.75%), indicating that its pretraining distribution or inference strategy does not transfer cleanly to ERCOT's extreme price regime. This is likely because TimesFM is built on a patch-based architecture (it splits the entire dataset into small chunks/patches and treats each patch as an individual token), as opposed to the next-token-prediction architecture of Chronos. The patch based architecture performs badly during extreme events that do not mirror any past chain of events.

The hybrid approach in which the foundation model's univariate forecast is appended as an additional input feature to XGBoost, which is then trained on the full 87-feature matrix plus the foundation forecast, produces the two most notable results. The Chronos hybrid achieves MAE of $12.13/MWh and RMSE of $93.98/MWh, the lowest values observed across all confirmed model configurations, while also achieving spike recall of 61.40% and spike precision of 50.43%. The Prophet hybrid, by contrast, improves on standalone Prophet's catastrophic MAE ($119.14/MWh → $48.36/MWh) but remains far behind the other hybrid, with spike precision of 14.16% despite high recall. The Prophet hybrid's failure to achieve comparable results is

consistent with Prophet's known weakness on non-stationary, heavy-tailed series (Taylor & Letham, 2018).

4.4 Classical Time Series

SARIMAX with daily seasonality (m=24) and the full exogenous regressor set achieves MAE of $18.45/MWh and RMSE of $124.31/MWh, with balanced spike recall (45.61%) and precision (38.81%). The higher RMSE relative to the GBT family is expected because SARIMAX assumes linear relationships between regressors and price, which does not work well at the spike tail where nonlinear interactions dominate.

Prophet performs poorly with MAE of $119.14/MWh. Prophet's additive decomposition into trend, seasonality, and regressor components is a poor fit for ERCOT prices, which have no stable trend and whose extreme values arise from threshold-crossing events rather than additive anomalies. The high spike recall (57.14%) alongside low precision (17.70%) suggests Prophet produces high variance forecasts that cross the $200 threshold frequently but are not accurate.

5. Secondary Findings

5.1 Sentiment Augmentation via GDELT

The contribution of news sentiment is measured by comparing a XGBoost model without the sentiment features against the baseline model we used in our previous comparisons. All other hyperparameters and the training procedure remain identical.

Configuration	MAE ($/MWh)	RMSE ($/MWh)	Spike Recall	Spike Precision
XGBoost (No sentiment)	$34.53/MWh	$122.57/MWh	50.53%	19.54%
XGBoost	$24.43/MWh	$112.46/MWh	57.02%	25.10%

GDELT sentiment reduces MAE by $10.10 (29.3%) and RMSE by $10.11 (8.2%). Spike precision improves by 5.56 percentage points and spike recall by 6.49 percentage points,

indicating that energy news tone carries information about the market conditions that precede price spikes.

6. Discussion

The GBT family dominates on MAE, with CatBoost achieving $13.75/MWh against the deep learning family's best of $24.42/MWh and the zero-shot foundation models' best of $14.72/MWh from Chronos. The exception is the Chronos hybrid, which achieves the best MAE and RMSE in the entire comparison by combining Chronos's temporal pattern recognition with XGBoost's ability to exploit the physical and sentiment feature set.

The deep learning results are the most poor. Both the LSTM and TFT underperform the XGBoost baseline on RMSE and spike metrics despite their architectural capacity for temporal modeling. This is consistent with expectations since deep learning models generally require millions/billions of datapoints to train on. The 80,000 row set that this project deals with is simply not enough for a deep learning architecture to train on. The TFT's zero spike recall is a particularly sharp failure, suggesting the model has not learnt the conditions that produce extreme prices even after training on nine years of data.

The sentiment ablation result, a 29.3% MAE improvement from augmenting the sentiment features, is the single largest contribution identified in the study. This is a strong finding, though it is worth noting that the ablation is on one model family and one test year. More testing needs to be done to solidify this claim.

7. Conclusion

This paper compares twelve forecasting architectures for ERCOT real-time electricity prices across five model families, evaluated on a shared 87-feature matrix spanning nine years of hourly observations. The results show that gradient boosted trees remain the most reliable family for this problem, with CatBoost and the Chronos hybrid achieving the strongest point estimates. Zero-shot foundation models show promise on typical hours but fail to generalize to spike conditions without feature augmentation. Classical time series and deep learning approaches underperform relative to tree-based methods, consistent with the nonlinear, heavy-tailed character of the ERCOT price distribution.

The most consequential individual finding is the GDELT sentiment ablation: removing news sentiment features degrades MAE by 29.3% on the 2024 test set, a larger effect than any single

architectural choice. This supports the inclusion of news sentiment as a standard component of electricity price forecasting pipelines, particularly for markets with the event-driven spike structure of ERCOT. Future work includes evaluating whether the GDELT improvement holds across model families and test years with different event compositions.

References

Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., ... & Wang, Y. (2024). Chronos: Learning the language of time series. arXiv preprint arXiv:2403.07815.

Das, A., Kong, W., Leach, A., Mathur, S., Sen, R., & Yu, R. (2024). Long-term forecasting with TiDE: Time-series dense encoder. arXiv preprint arXiv:2304.08424.

Potomac Economics. (2021). 2020 State of the market report for the ERCOT electricity markets. Independent Market Monitor for ERCOT.

Taylor, S. J., & Letham, B. (2018). Forecasting at scale. The American Statistician, 72(1), 37–45.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785–794).

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., ... & Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. Advances in Neural Information Processing Systems, 30.

Duan, T., Avati, A., Ding, D. Y., Thai, K. K., Basu, S., Ng, A., & Schuler, A. (2020). NGBoost: Natural gradient boosting for probabilistic prediction. In International Conference on Machine Learning (pp. 2690–2700). PMLR.